

Homework #2

Math 104: Intro to Analysis

Summer 2026

UC Berkeley

Donald Aingworth IV

July 10, 2026

Contents

1	Problem 1	2
1.1	Solution (a)	3
1.2	Solution (b)	4
1.3	Solution (c)	5
2	Problem 2	6
2.1	Solution (a)	7
2.2	Solution (b)	8
2.3	Solution (c)	9
2.4	Solution (d)	10
3	Problem 3	11
3.1	Solution (a)	12
3.2	Solution (b)	13
3.3	Solution (c)	14
4	Problem 4	15
4.1	Solution (a)	16
4.2	Solution (b)	17
4.3	Solution (c)	18
4.4	Solution (d)	19
4.5	Solution (e)	20
4.6	Solution (f)	21

1 Problem 1

Let (X, d) be a metric space. A subset $E \subset X$ is called *dense* in E if, for any open $O \subset X$, we have $O \cap E \neq \emptyset$. A metric space (X, d) is called *separable* if it has a countable dense subset.

- (a) Show that $\mathbb{Q}$ is dense in $\mathbb{R}$, and conclude that $\mathbb{R}$ is separable.
- (b) Show that a subset E is dense in X if and only if $\overline{E} = X$.
- (c) By following these steps, prove that any compact metric space (X, d) is separable:
 - (i) Using problem 5(c) on homework 2, show that for each $n \in \mathbb{N}$, the following is true: there is a finite set $S_n \subset X$ so that every point of X is within distance $\frac{1}{n}$ of some point of S_n .
 - (ii) Let $S = \bigcup_{n=1}^{\infty} S_n$. Show that S is countable.
 - (iii) Show that every point of X is either in S or a limit point of S . Conclude that S is dense in X , and hence that X is separable.

1.1 Solution (a)

Alexis's Proposition: For any $a, b \in \mathbb{R}$ where $a < b$, there exists some $\frac{m}{n} \in \mathbb{Q}$ such that $a < \frac{m}{n} < b$.

Proof. Let $a, b \in \mathbb{R}$ where $a < b$. Since $a < b$, $b - a > 0$. By the Archimedean property, there exists some $n \in \mathbb{N}$ such that $n(b - a) > 1$. Distribute and add na to both sides to find $nb > na + 1$. Also by the Archimedean property, there exists some natural numbers m_1 and m_2 such that $m_1 > na$ and $m_2 > -na$. The latter means that $na > -m_2$. Therefore, $m_1 > na > -m_2$. Since m_1 and $-m_2$ are both integers, there exists some integer m such that $-m_2 < m < m_1$ and $m - 1 \leq na < m$. This is equivalent to $m \leq na + 1 < m + 1$. Combine all of these and we find that $na < m \leq na + 1 < nb$. Shortening this, $na < m < nb$. Divide this all by n to find $a < \frac{m}{n} < b$. Since n is a natural number and m is an integer, $\frac{m}{n}$ is a rational number. Ergo, if $a, b \in \mathbb{R}$ and $a < b$, there exists some $\frac{m}{n} \in \mathbb{Q}$ such that $a < \frac{m}{n} < b$. OEΔ

Now we prove $\mathbb{Q}$ is dense in $\mathbb{R}$ with the usual metric.

Proof. Take any set O open in $\mathbb{R}$. This means that for any $o \in O$, there exists some $r > 0$ such that there exists an open ball $B_r(o)$ that is a subset of O . We define this open ball as $B_r(o) = \left\{ x \in \mathbb{R} \mid |x - o| < r \right\}$.

Since $r > 0$, this means $r > -r$ and therefore $o + r > o - r$. By Alexis's Proposition, there exists some rational number $\frac{m}{n}$ such that $o - r < \frac{m}{n} < o + r$. Subtract o from all sides to find $-r < \frac{m}{n} - o < r$. By the definition of the absolute value, $\left| \frac{m}{n} - o \right| < r$. Therefore, $\frac{m}{n} \in B_r(o)$ and by extension $\frac{m}{n} \in O$. Therefore, $\frac{m}{n} \in \mathbb{Q} \cap O$. Therefore, OEΔ

1.2 Solution (b)

1.3 Solution (c)

2 Problem 2

Using **only** the definitions of $\lim$, $\limsup$, and $\liminf$, prove the following statements.

(a) $\lim_{n \rightarrow \infty} [\sqrt{n+1} - \sqrt{n}] = 0.$ *Hint:* Use difference of squares.

(b) $\limsup_{n \rightarrow \infty} (-2)^n = \infty.$

(c) $\limsup_{n \rightarrow \infty} [-\sqrt{n}] = -\infty.$

(d) $\liminf_{n \rightarrow \infty} \left[\frac{1}{n} + (1 + (-1)^n)n \right] = 0.$

2.1 Solution (a)

2.2 Solution (b)

2.3 Solution (c)

2.4 Solution (d)

3 Problem 3

Let $\{s_n\}$ be an arbitrary sequence in $\mathbb{R}$.

(a) Let $s \in \mathbb{R}$. Show that $\limsup_{n \rightarrow \infty} s_n = s$ if and only if the following two conditions hold:

- For all $\epsilon > 0$, there is $N \in \mathbb{N}$ so that $s_n < s + \epsilon$ for all $n \geq N$.
- For all $\epsilon > 0$ and for all $N \in \mathbb{N}$, there exists $n \geq N$ so that $s_n > s - \epsilon$.

(b) Show that some subsequence of $\{s_n\}$ converges (in the sense of extended real numbers, if necessary) to $\limsup_{n \rightarrow \infty} s_n$ and some subsequence of $\{s_n\}$ converges to $\liminf_{n \rightarrow \infty} s_n$.

(c) Define

$$E = \{x \in [-\infty, \infty] : \text{some subsequence } \{s_{n_k}\} \text{ of } \{s_n\} \text{ converges to } x\}.$$

Show that $\limsup_{n \rightarrow \infty} s_n = \sup E$ and $\liminf_{n \rightarrow \infty} s_n = \inf E$.

Hint: what happens to $\limsup$ and $\liminf$ on subsequences?

3.1 Solution (a)

3.2 Solution (b)

3.3 Solution (c)

4 Problem 4

Let $a > 1$. In this problem, we will define what it means to write a^r for a positive *irrational* number r . We first define exponentiation for positive *integer* exponents in the usual way, via repeated multiplication. We know this satisfies some properties, for example

$$a^{r+s} = a^r a^s \quad (*)$$

$$(ab)^r = a^r b^r \quad (\Delta)$$

for $a, b > 0$ and $r, s > 0$ integers. Now, we proved in class that for any positive integer q , there is a *unique* positive number whose q th power is a , which we denote $\sqrt[q]{a}$. For a positive rational number $r = \frac{p}{q}$, we define $a^r = \sqrt[q]{a^p}$.

- (a) Show that this definition makes sense, i.e. is independent of the representation of r as a fraction. That is, show that if rational numbers $\frac{p}{q} = \frac{r}{s}$ then $\sqrt[q]{a^p} = \sqrt[s]{a^r}$. *Hint:* use lowest terms and the uniqueness of q th roots.
- (b) Show that properties $(*)$ and (Δ) above still hold for positive rational numbers. *Hint:* find a common denominator and use uniqueness again.
- (c) For positive rational $r < s < t$, show that $a^r < a^s$ and that

$$0 < a^s - a^r = a^r (a^{s-r} - 1).$$

Conclude that for any positive rational r, s and any rational t larger than both r and s , we have

$$|a^r - a^s| < a^t (a^{|r-s|} - 1).$$

- (d) Now let r be a positive *irrational* number. We know there is a sequence of rational numbers q_k that converge to r (for example, the decimal expansion of r). We define

$$a^r = \lim_{k \rightarrow \infty} a^{q_k}.$$

Show that the sequence $\{a^{q_k}\}$ is Cauchy, and conclude that this limit exists.

Hint: Use part (c) and Rudin Theorem 3.20(d).

- (e) Show that the choice of sequence $\{q_k\}$ is irrelevant; that is, show that if $\{p_k\}$ and $\{q_k\}$ are sequences of rational numbers converging to r , then

$$\lim_{k \rightarrow \infty} a^{p_k} = \lim_{k \rightarrow \infty} a^{q_k}.$$

- (f) Using limit laws, prove that properties $(*)$ and (Δ) continue to hold for positive irrational exponents.

Remark: For $r < 0$ we then define $a^r = \frac{1}{a^{-r}}$, and for $0 < a < 1$ we define $a^r = \left(\frac{1}{a}\right)^{-r}$. It is then a matter of checking cases to verify that all the properties of exponentiation we want are indeed true.

4.1 Solution (a)

Billie's Proposition: If $\frac{p}{q} = \frac{r}{s}$ and $\frac{p}{q}$ is in lowest terms but $\frac{r}{s}$ is not, there exists some nonzero integer t such that $r = pt$ and $s = qt$.

Proof. Let $\frac{p}{q} = \frac{r}{s}$ and $\frac{p}{q}$ is in lowest terms but $\frac{r}{s}$ is not. Multiply by to find that $ps = qr$.

Divide by p to find that $s = \frac{qr}{p}$. Since s is a nonzero integer, p divides qr . Since $\frac{p}{q}$ is in lowest terms, p does not divide q , so p must divide r . Therefore, there is some nonzero integer t such that $p = rt$.

Substitute in our value of p into the result of the cross multiplication to find $rst = qr$. Divide both sides by r to find that $q = st$. OEΔ

Proof. Let $x = \sqrt[q]{a^p}$ and $y = \sqrt[s]{a^r}$, and suppose that $\frac{p}{q} = \frac{r}{s}$ and that $\frac{p}{q}$ is in lowest terms while $\frac{r}{s}$ is not. This would mean that $x = a^{\frac{p}{q}}$ and $y = a^{\frac{r}{s}}$. Take x to the power of q and y to the power of s to find $x^q = a^p$ and $y^s = a^r$. Take the former to the power of r and the latter to the power of p to find $x^{qr} = a^{pr}$ and $y^{ps} = a^{pr}$. Transistively, this means $x^{qr} = y^{ps}$.

By Billie's Proposition, since $\frac{p}{q}$ is in lowest terms while $\frac{r}{s}$ is not, there exists some nonzero integer t such that $r = pt$ and $s = qt$. Substitute these into our equation to find that $x^{qpt} = y^{qpt}$.

$$x^{qr} = y^{sp} \tag{1}$$

$$x = y \tag{2}$$

$$(x^{qr})^{sp} = (y^{sp})^{qr} \tag{3}$$

$$x^{qr} = (y^{sp})^{\frac{q}{p} \cdot \frac{r}{s}} \tag{4}$$

OEΔ

4.2 Solution (b)

4.3 Solution (c)

4.4 Solution (d)

4.5 Solution (e)

4.6 Solution (f)